SHIVANSH GARG

ACADEMIC DETAILS

Year	Degree / Board	Institute	GPA / Marks(%)
---	M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63
2024	B.E in Computer Science and Engineering	UIET, Panjab University	9.37
2020	CBSE	Shaheed Bhai Diala Ji Public School	94.2%
2018	CBSE	Sacred Heart Convent School	97.8%

QUALIFYING EXAMS

- **Graduate Aptitude Test in Engineering (GATE) Rank:** 75

SCHOLASTIC ACHIEVEMENTS

- **Department Rank 3:** Current rank in M.Tech (Computer Science & Engineering), IIT Delhi.

IIT DELHI THESIS

Title: Atrial Fibrillation Detection Using PPG Signals in Wearable Devices
Supervisor: Prof. Rahul Garg
Description: Developing **machine learning** algorithms to enhance **Atrial Fibrillation detection** using **PPG signals from wearable devices**. This research focuses on **minimizing false positives** by distinguishing cardiac events from motion artifacts and investigating PPG-derived features to identify **subtypes of AF**, thereby improving remote cardiac monitoring precision.

PROJECTS

- **P-Wave Detection in Electrocardiograms (Prof. Rahul Garg)** *Minor Project* :
  - Performed a **systematic literature review** of **30** studies on automated P-wave detection covering signal processing, **machine learning, and deep learning approaches**.
  - **Implemented and validated algorithms** and benchmarked the **NeuroKit2 library**, assessing overall **reproducibility.**
  - Conducted **comparative evaluation** of detection methods on **ECG databases**, analyzing performance and limitations.

- **Visual Question Answering with Multi-Modal Transformers (Prof. Parag Singla)**:
  - Built a multi-modal transformer with a **ResNet image encoder** and **transformer text encoder**.
  - Used **cross-attention** fusion to improve **visual-text reasoning** on the CLEVR dataset.
  - Improved performance using **BERT embeddings and focal loss**, and validated generalization through **zero-shot**.

- **Neural Cleanse: Backdoor Attack Detection in Deep Neural Networks (Prof. Huzur Saran)**:
  - Implemented and evaluated **blind backdoor attacks** on **deep neural networks** to study stealthy adversarial behavior.
  - Implemented defenses with **differential privacy** and certified robustness, analyzing effectiveness against attacks.

- **Text and Image Classification using Naïve Bayes and SVM (Prof. Parag Singla)**:
  - Implemented **Naïve Bayes from scratch** with preprocessing and bigram based text features for text classification.
  - Built and **evaluated SVM** with linear and Gaussian kernels using CVXOPT and scikit-learn for image classification.

- **Representation Learning with Neural Networks (Prof. Rohan Paul)**:
  - Implemented a custom **CNN** for 10-class bird classification with **Grad-CAM**-based feature interpretability.
  - Developed a **Variational Autoencoder (VAE)** for unsupervised representation learning and a from-scratch Gaussian Mixture Model (GMM) for few-shot classification.

- **Search-Based Agent for ASR Error Correction (Prof. Rohan Paul)**:
  - Developed a **local search agent** to correct phonetic and missing word errors in **speech recognition** output.

TECHNICAL SKILLS

- **Programming Languages :** C/C++, Python, SQL
- **Libraries & Frameworks :** NumPy, Pandas, PyTorch, scikit-learn, Neurokit2, wfdb, SUMO, NetEdit

TEACHING ASSISTANTSHIP

- Teaching Assistant, **Principles of Artificial Intelligence (COL333)** | Prof. Mausam | Diwali 2025
- Teaching Assistant, **Analysis and Design of Algorithms (COL351)** | Prof. Keerti Chaudary | Holi 2025
- Teaching Assistant, **Analysis and Design of Algorithms (COL351)** | Prof. Rohit Vaish | Diwali 2024

SHIVANSH GARG

IIT COURSE

Degree	Institute	CGPA	Dept. Rank
M.Tech in Computer Science & Engineering	Indian Institute of Technology Delhi	8.63	---

COURSES DONE

Advanced Data Structures, Artificial Intelligence, Intro. To Logic & Funct. Prog., Software Systems Laboratory, Teaching Practicum In Computer Science And Engineering, Machine Learning, Resource Management In Computer Systems, Networks & System Security, Minor Project, Teaching Practicum In Computer Science And Engineering